

GPT-6 OLI Open-Loop Trace

Three independent observations from episode 0 of 0811_candy_v21 | 2026-09-12

Objective

Evaluate whether GPT-6 selects the correct task phase, wrist translation, wrist rotation, and hand command while producing a 50 x 66 keypoint action chunk. No action was published to a robot.

Task

Goal: Pick up the purple candy and place it in the right section of the snack tray with right arm.

Runtime contract

- Model: gpt-6-astra; reasoning effort: medium; image detail: high.
- Each frame is an independent call (Action call 1/50); no synthetic execution history.
- Wrist/head translation output: base-frame relative dx/dy/dz.
- Wrist rotation output: wrist-local relative roll/pitch/yaw in degrees.
- Output: 50-step, 66-D trajectory at the policy boundary; no Operator or MROS publishing.

Offline-data caveat

The source parquet stores observation.base_pose translation as [0, 0, 0]. It does not represent live base height. The dataset also lacks foot-pose observations, so the test harness supplies zero-position identity-pose foot placeholders. Live inference obtains these values from MROS.

Complete system prompt

Part 1

You control an OLI humanoid through relative wrist and head motions. At every turn you receive head, left-wrist, and right-wrist camera images, the current absolute poses, and both hand states. Use exactly one `control_oli` tool call. Omitted targets hold their current values.

Infer the current task phase from the images, robot state, and prior turns. Continue from the observed state; do not restart a fixed grasping sequence or infer the phase from finger closure alone. Make one small, deliberate move. Each hand target contains its six native finger channels in this order: [thumb flexion, thumb lateral splay, index, middle, ring, little]. Useful reference profiles are open [0, 98, 0, 0, 0, 0], normal grasp [30, 92, 40, 98, 98, 98], and firm grasp [58, 92, 58, 98, 98, 98]. Values between profiles provide partial closure. Channel 1 controls lateral thumb splay rather than finger flexion, so its open value intentionally stays near 98. Omit an unused hand to hold all six current channels. Keep the unused wrist and head still unless moving them is necessary. Position units are metres in the robot base frame: +x points forward from the robot, +y points to the robot's left, and +z points upward. Wrist and head $dx/dy/dz$ commands are relative displacements in that frame. Wrist rotation commands are relative to a right-handed hand frame: local +x points from wrist to fingers, local +y lies laterally across the palm, and local +z is the outward palm normal. Roll twists the wrist about +x, pitch moves the fingers up/down about +y, and yaw moves the fingers left/right about +z. Positive angles follow the right-hand rule. Use wrist rotation when the visible finger or palm alignment requires it. Include a short note explaining the visible evidence and requested move. Commands are bounded by the runtime; use the next observation to verify what was executed.

Part 2

Control tool supplied to GPT

Readable equivalent of the JSON schema embedded in every request

```
control_oli({
  "targets": {
    "left_wrist"?: {
      "dx"?: number,
      "dy"?: number,
      "dz"?: number,
      "rotation_delta_rpy_deg"?: [number, number, number]
    },
    "right_wrist"?: { same schema as left_wrist },
    "head"?: { "dx"?: number, "dy"?: number, "dz"?: number },
    "left_hand"?: [number x 6, each in 0..100],
    "right_hand"?: [number x 6, each in 0..100]
  },
  "note": string
})
```

```
Hand order:
[thumb_flex, thumb_splay, index, middle, ring, little]
```

```
Reference profiles:
open      [0, 98, 0, 0, 0, 0]
normal grasp [30, 92, 40, 98, 98, 98]
firm grasp  [58, 92, 58, 98, 98, 98]
```

```
Omitted fields hold their current values.
```

Frame 40

Independent open-loop call | latency 8.66 s | input/output tokens 2434/157

head



left wrist



right wrist



Complete text observation

Current OLI observation.
Instruction: Pick up the purple candy and place it in the right section of the snack tray with right arm.
Action call: 1/50.
Max wrist displacement per call: 0.080 m.
Max head displacement per call: 0.030 m.
Max relative rotation per call: 15.0 degrees.
Workspace xyz bounds: [[-0.3 , 0.8],
[-0.65, 0.65],
[-0.3 , 1.5]]
left_wrist xyz in base frame: [0.16357, 0.21532, 0.12475]
right_wrist xyz in base frame: [0.22993, -0.17302, 0.05636]
head xyz in base frame: [0.06312, 0.00871, 0.58252]
left_foot xyz in base frame: [0., 0., 0.]
right_foot xyz in base frame: [0., 0., 0.]
base xyz in world frame: [0., 0., 0.]
left_hand [thumb_flex, thumb_splay, index, middle, ring, little]: [2., 97., 2., 2., 2., 2.]
right_hand [thumb_flex, thumb_splay, index, middle, ring, little]: [29., 92., 39., 97., 97., 97.]

Analysis

Direction +y is useful, but GPT omits +x, descent, and wrist rotation. It opens the hand while GT is already transitioning toward a grasp.

GPT tool call

```
{
  "targets": {
    "right_wrist": {
      "dy": 0.025
    },
    "right_hand": [
      0,
      98,
      0,
      0,
      0,
      0
    ]
  },
  "note": "The purple candy is on the tabletop just left of the right hand, not yet grasped. Open the right hand and shift slightly left to begin aligning over it; hold the left arm and head still."
}
```

Converted endpoint vs GT

66-D endpoint
right wrist xyz: [0.22992999851703644, -0.14801999926567078, 0.056359998881816864]
head xyz: [0.06312000006437302, 0.008709999732673168, 0.5825200080871582]
hand wire [L0,R0,...]: [2.0, 0.0, 97.0, 98.0, 2.0, 0.0, 2.0, 0.0, 2.0, 0.0, 2.0, 0.0]

Ground truth
right wrist delta: [0.02143, 0.01279, -0.00739]
right wrist local RPY deg: [2.54, -1.26, 6.38]
hand wire [L0,R0,...]: [0.0, 30.0, 98.0, 92.0, 0.0, 40.0, 0.0, 98.0, 0.0, 98.0, 0.0, 98.0]

Frame 120

Independent open-loop call | latency 9.58 s | input/output tokens 2434/108

head



left wrist



right wrist



Complete text observation

Current OLI observation.
Instruction: Pick up the purple candy and place it in the right section of the snack tray with right arm.
Action call: 1/50.
Max wrist displacement per call: 0.080 m.
Max head displacement per call: 0.030 m.
Max relative rotation per call: 15.0 degrees.
Workspace xyz bounds: [[-0.3 , 0.8],
[-0.65, 0.65],
[-0.3 , 1.5]]
left_wrist xyz in base frame: [0.16952, 0.2149 , 0.12313]
right_wrist xyz in base frame: [0.25183, -0.17172, 0.01647]
head xyz in base frame: [0.06703, 0.00812, 0.58193]
left_foot xyz in base frame: [0., 0., 0.]
right_foot xyz in base frame: [0., 0., 0.]
base xyz in world frame: [0., 0., 0.]
left_hand [thumb_flex, thumb_splay, index, middle, ring, little]: [2., 97., 2., 2., 2., 2.]
right_hand [thumb_flex, thumb_splay, index, middle, ring, little]: [55., 92., 54., 97., 97., 97.]

Analysis

Grip retention is correct, but GPT chooses lift. GT mainly moves +y with a small descent and a 8.3-degree local rotation. Single-frame phase remains ambiguous.

GPT tool call

```
{
  "targets": {
    "right_wrist": {
      "dz": 0.035
    }
  },
  "note": "The purple candy is beneath the curled right fingers beside the tray. Lift the right wrist slightly while holding the current grip to check that the candy is secured and clear of the table."
}
```

Converted endpoint vs GT

66-D endpoint
right_wrist xyz: [0.25183001160621643, -0.1717199981212616, 0.05147000029683113]
head xyz: [0.06702999770641327, 0.00812000036239624, 0.5819299817085266]
hand_wire [L0,R0,...]: [2.0, 55.0, 97.0, 92.0, 2.0, 54.0, 2.0, 97.0, 2.0, 97.0, 2.0, 97.0]

Ground truth
right_wrist_delta: [0.00444, 0.0403, -0.01042]
right_wrist_local_RPY_deg: [5.26, -0.98, -6.33]
hand_wire [L0,R0,...]: [0.0, 58.0, 98.0, 92.0, 0.0, 58.0, 0.0, 98.0, 0.0, 98.0, 0.0, 98.0]

head



left wrist



right wrist



Complete text observation

Current OLI observation.
Instruction: Pick up the purple candy and place it in the right section of the snack tray with right arm.
Action call: 1/50.
Max wrist displacement per call: 0.080 m.
Max head displacement per call: 0.030 m.
Max relative rotation per call: 15.0 degrees.
Workspace xyz bounds: [[-0.3 , 0.8],
[-0.65, 0.65],
[-0.3 , 1.5]]
left_wrist xyz in base frame: [0.15833, 0.21148, 0.12336]
right_wrist xyz in base frame: [0.3022 , -0.10138, 0.0649]
head xyz in base frame: [0.06671, 0.00872, 0.58197]
left_foot xyz in base frame: [0., 0., 0.]
right_foot xyz in base frame: [0., 0., 0.]
base xyz in world frame: [0., 0., 0.]
left_hand [thumb_flex, thumb_splay, index, middle, ring, little]: [2., 97., 2., 2., 2., 2.]
right_hand [thumb_flex, thumb_splay, index, middle, ring, little]: [30., 92., 40., 97., 97., 97.]

GPT tool call

```
{
  "targets": {
    "right_wrist": {
      "dy": 0.04,
      "dz": 0.025
    },
    "right_hand": [
      0,
      98,
      0,
      0,
      0,
      0
    ]
  },
  "note": "The purple candy is resting in the tray, left of the empty right hand. Open the hand and shift slightly left and upward to begin aligning above it with clearance."
}
```

Converted endpoint vs GT

66-D endpoint
right wrist xyz: [0.30219998955726624, -0.06137999892234802, 0.08990000188350677]
head xyz: [0.06671000272035599, 0.00872000027447939, 0.5819699764251709]
hand wire [L0,R0,...]: [2.0, 0.0, 97.0, 98.0, 2.0, 0.0, 2.0, 0.0, 2.0, 0.0, 2.0, 0.0]

Ground truth
right wrist delta: [-0.00715, 0.02118, 0.01294]
right wrist local RPY deg: [-0.85, 1.35, -14.65]
hand wire [L0,R0,...]: [0.0, 0.0, 98.0, 98.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]

Analysis

Best match. GPT and GT both command +y/+z and an open right hand. GPT omits the small -x component and the large -14.65-degree local yaw.

Findings and limitations

1. Removing the hard-coded open → approach → descend → close → lift recipe reduced phase bias, but a single image remains ambiguous around grasp/transport transitions.
2. Frame 199 matches GT direction and hand state most closely. Frame 120 still chooses lift instead of the GT lateral transfer.
3. GPT omitted wrist rotation in all three calls even when GT applies 7–15 degrees. Rotation guidance remains the clearest policy weakness.
4. The corrected translation conversion is $\text{target_xyz} = \text{current_xyz} + \text{delta_xyz}$ because both observation and requested displacement are in the robot base frame.
5. This dataset cannot validate mobile-base translation: its stored base translation is identically zero. A live MROS trace or a dataset containing measured base translation is required for that test.

Trace artifacts

```
frame_40_request.json
frame_40_response.json
frame_40_summary.json
frame_120_request.json
frame_120_response.json
frame_120_summary.json
frame_199_request.json
frame_199_response.json
frame_199_summary.json
```

The API returned encrypted reasoning payloads with empty public summary/content. This report therefore includes every observable prompt and tool output but does not claim access to hidden chain-of-thought.